

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Final Examination of Semester 1
Subject: 353152 Electronic Circuits I
Date: 15 December 2014

Year: 2014
Section: 5 - 6
Time: 13.00-15.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Textbooks and Dictionaries are **NOT** allowed.
4. A calculator is allowed.
5. Electronic communication devices are **NOT** allowed in the examination room.
6. The examination has 11 pages (including this page), 2 parts (55 questions) and a total score of 150 points.
7. Write all your answer on this examination sheet.

วิทยาลัยเทคโนโลยีอุตสาหกรรม

Part 1 (100 points) Multiple choices

Instruction: Mark the correct answer for the following questions in the provided answer sheet.

- From Figure 1, determine R_C .
 (A) $1\text{ k}\Omega$ (B) $2.2\text{ k}\Omega$ (C) $4.8\text{ k}\Omega$ (D) $6\text{ k}\Omega$
- From Figure 1, determine R_B .
 (A) $356\text{ k}\Omega$ (B) 0.356 (C) $452\text{ k}\Omega$ (D) $384\text{ k}\Omega$

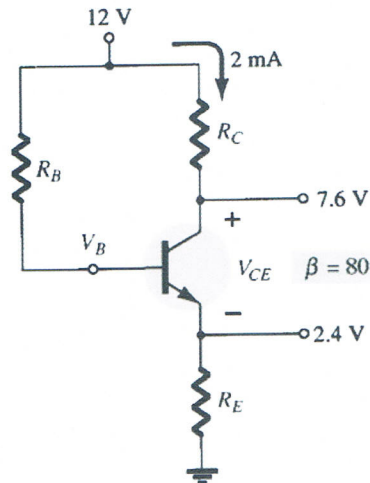


Figure 1.

- Which of the following voltages must have a positive level (value) in any PNP bias circuit?
 (A) V_{BE} (B) V_{CE} (C) V_{BC} (D) None of above

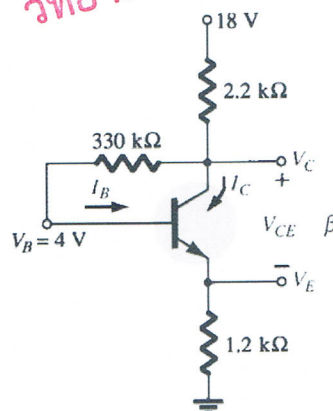


Figure 2.

- From Figure 2, find the current I_E .
 (A) 23.78 mA (B) 2.37 mA (C) 3.41 mA (D) 2.75 mA
- From Figure 2, find the current V_C .
 (A) 9.87 V (B) 12.78 V (C) 11.95 V (D) 10.5 V
- For the most dc configurations which of the following law is used?
 (A) KCL (B) KVL (C) Ohm's law (D) Thevenin

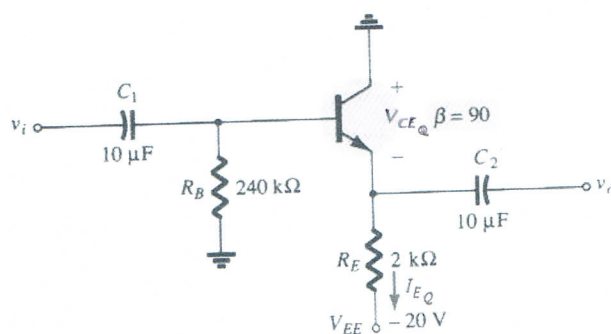


Figure 3.

7. From Figure 3, find the current I_B .
 (A) 80.16 μA (B) 80.16 mA (C) used 45.73 μA (D) 45.73 mA
8. From Figure 3, which of following equations is for finding V_{CE} ?
 (A) $V_{CE} = V_{EE} - 0.7 - I_E R_E$ (B) $V_{CE} = V_{EE} - I_E R_E$
 (C) $V_{CE} = V_{EE} - I_E R_E - I_B R_B$ (D) $V_{CE} = V_{EE} - I_E R_E - 0.7 - I_B R_B$
9. What is the equation when the transistor is operating in saturation mode?
 (A) $I_C = \frac{V_{CC}}{R_C}$ (B) $V_{CE} = V_{CC}$
 (C) $I_B = \frac{V_{CC} - V_{BE}}{R_B}$ (D) $V_{CE} = V_{EE} - I_E R_E$
10. For the most dc configuration, the analysis will start by _____.
 (A) base current (B) emitter current
 (C) collector current (D) emitter-collector voltage
11. Which of following circuits, base is connected directly to the ground?
 (A) fixed-bias (B) emitter-bias
 (C) common base (D) collector feedback
12. In an emitter-bias configuration, the larger the resistance R_E , the _____ the stability factor, and the _____ stable is the system.
 (A) lower, less (B) more, more
 (C) more, more (D) lower, more
13. In a transistor-switching network, the level of the resistance between the collector and emitter is _____ at the cut-off and is _____ at the saturation.
 (A) low, low (B) low, high (C) high, high (D) high, low
14. For the typical transistor amplifier in the active region, V_{CE} is usually about _____ % to _____ % of V_{CC} .
 (A) 0, 100 (B) 25, 75 (C) 45, 55 (D) 50, 50
15. Which of the following is assumed in the approximate analysis of a voltage divider circuit?
 (A) I_B is essentially zero ampere (B) $\beta R_E \geq 10 R_2$
 (C) R_1 and R_2 are considered to be series elements. (D) All of the above

16. _____ is the primary difference between the exact and approximate techniques used in the analysis of a voltage divider circuit.
- (A) Thevenin voltage V_{th} (B) Thevenin resistance R_{Th}
 (C) Base voltage V_B (D) R_C
17. For a transistor operating in the saturation region, the collector current I_C is at its _____ and the collector-emitter voltage V_{CE} is to the _____.
- (A) minimum, left of the V_{CEsat} line (B) minimum, right of the V_{CEsat} line
 (C) maximum, left of the V_{CEsat} line (D) maximum, right of the V_{CEsat} line
18. The _____ the stability factor, the _____ sensitive the network is to variations in that parameter.
- (A) higher, more (B) higher, less
 (C) lower, more (D) None of the above
19. Which one of the stability factors is becoming the least concern in the design process?
- (A) $S(I_{CO})$ (B) $S(V_{BE})$ (C) $S(\beta)$ (D) $S(V_{CE})$
20. In a fixed-bias circuit, which one of the stability factors overrides the other factors?
- (A) $S(I_{CO})$ (B) $S(V_{BE})$ (C) $S(\beta)$ (D) Undefined
21. In a transistor-switching network, the operating point switches from _____ to _____ regions along the load line.
- (A) cutoff, active (B) cutoff, saturation
 (C) active, saturation (D) None of the above
22. _____ is the least stabilized circuit.
- (A) Fixed bias (B) Emitter-stabilized bias
 (C) Voltage divider (D) Voltage feedback
23. _____ is less dependent on the transistor beta.
- (A) Fixed bias (B) Emitter-stabilized bias
 (C) Voltage divider (D) Voltage feedback
24. The Thevenin equivalent network is used in the analysis of the _____ circuit.
- (A) Fixed bias (B) Emitter-stabilized bias
 (C) Voltage divider (D) Voltage feedback
25. The emitter resistor in an emitter-stabilized bias circuit appears to be _____ in the base circuit.
- (A) larger (B) smaller
 (C) the same (D) None of the above
26. Which one of the following will affect the level of both current gain β and leakage current I_{CO} ?
- (A) Humidity (B) Speed (C) Temperature (D) Pressure
27. In any amplifier employing a transistor, the collector current I_C is sensitive to _____.
- (A) β (B) V_{BE} (C) I_{CO} (D) V_{CC}

28. Calculate I_E from Figure 4.

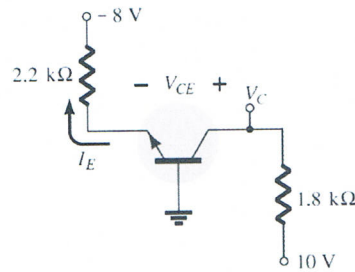


Figure 4.

- (A) 3.63 mA (B) 3.32 mA (C) 4.22 mA (D) 1.22 mA
29. In order to isolate the ac levels from the dc analysis in a transistor network, the capacitor can be replaced by _____.
- (A) an open circuit equivalent (B) a short circuit equivalent
(C) a voltage source (D) a current source

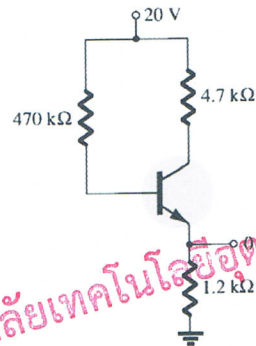


Figure 5.

30. From Figure 5, the measurement of $V_E=0$ V does not function correctly. What could NOT be the problem?
- (A) Open-circuit in the base circuit.
(B) Bad connection of emitter terminal.
(C) Damaged transistor.
(D) Short base-emitter junction.

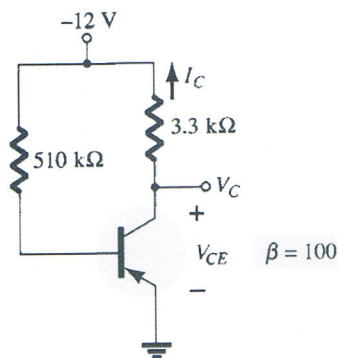


Figure 6.

31. From Figure 6, determine V_{CE} .
- (A) -7.26 V (B) -4.69 V (C) 4.69 V (D) 7.26 V

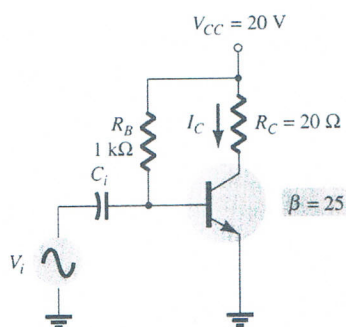


Figure 7.

32. From Figure 7, what is value of $P_o(AC)$ if $I_c = 250$ mA peak.
 (A) 1.25 mW (B) 1.25 W (C) 0.625 mW (D) 0.625 W
33. How many transistors must be used in a class B power amplifier to obtain the output for the full cycle of the signal?
 (A) 0 (B) 1 (C) 2 (D) 3
34. What is the disadvantage of the push-pull circuit?
 (A) Low gain (B) Low efficiency
 (C) Too expensive (D) Cross over distortion
35. What is efficiency of class A amplifier?
 (A) $\% \eta = \frac{P_o(dc)}{P_i(dc)} \times 100\%$ (B) $\% \eta = \frac{P_o(ac)}{P_i(dc)} \times 100\%$
 (C) $\% \eta = \frac{P_o(dc)}{P_i(ac)} \times 100\%$ (D) $\% \eta = \frac{P_o(ac)}{P_i(ac)} \times 100\%$
36. A heat sink provides _____ thermal resistance between case and air.
 (A) a high (B) a low
 (C) the same (D) None of the above
37. A _____ power amplifier is limited to use at one fixed frequency.
 (A) class A (B) class B or AB
 (C) class C (D) class D
38. Which of following transistor operating mode represents the short circuit?
 (A) cut-off (B) saturation (C) active (D) open
39. Which class of power amplifiers that conducts 180° of input?
 (A) class A (B) class B (C) class AB (D) class C
40. Which class of power amplifiers has Q-point below the cutoff point?
 (A) class A (B) class B (C) class AB (D) class C
41. Which type of following power amplifiers has the highest efficiency?
 (A) class A (B) class B (C) class AB (D) class C
42. How can the class A amplifier efficiency be improved from 25%?
 (A) Increasing input current. (B) Reducing input current.
 (C) Adding transformer at input side. (D) adding transformer at output side.

43. What is the benefit of using Darlington pair and feedback pair transistors for push-pull operation?
- (A) Reducing output swing. (B) Increasing output power.
(C) Increasing input current. (D) Reducing distortion.
44. Class D operation can achieve power efficiency of over ____.
- (A) 90% (B) 78.5% (C) 50% (D) 25%
45. Which type of amplifier uses pulse (digital) signals in its operation?
- (A) Class A (B) Class B or AB
(C) Class C (D) Class D
46. What is the value of I_{DC} of class B amplifier if peak output current $I_L(p)$ is 1A.
- (A) 0.318 A (B) 0.636 A (C) 0.707 A (D) 1.414 A
47. What is the value of $P_o(AC)$ of class B amplifier if $V_L(p-p) = 10$ V and $R_L = 2$ ohms.
- (A) 50 W (B) 25 W (C) 6.25 W (D) 5.25 W
48. What is the purpose of using push-pull circuit for class B amplifier?
- (A) to obtain full cycle (B) to reduce the distortion
(C) to increase speed of signal (D) to reduce the cost of amplifier
49. Calculate the delay time in a transistor switching network if t_{on} is 56 ns, $t_f = 14$ ns, and $t_r = 20$ ns.
- (A) 70 ns (B) 42 ns (C) 36 ns (D) 34 ns
50. Which of the following is the application of a transistor?
- (A) Switching (B) Motors
(C) Transformers (D) Fluorescents

Part 2 (50 points) Calculation

Instruction: Show the mathematical expression and answers of following problems.

1. From Figure 8, calculate the input power $P_i(\text{dc})$ and output power $P_o(\text{ac})$ of the circuit. The input signal results in a base current of 5mA_{RMS} , where $\beta = 80$ and $V_{CC} = 20\text{ V}$.

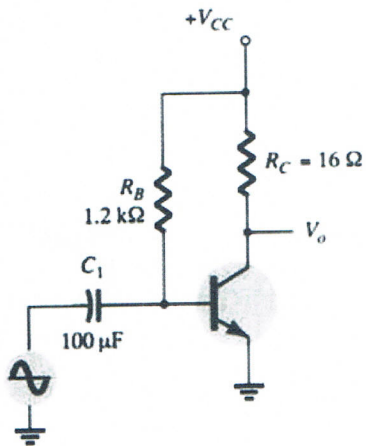


Figure 8.

2. From Figure 9, determine the level of currents I_E and I_B .

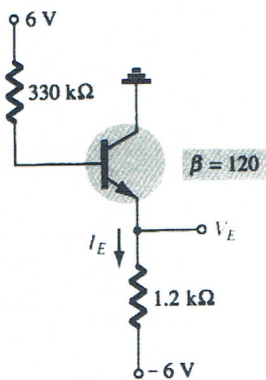


Figure 9.

3. From Figure 10, determine I_B , V_B , and V_E with $\beta = 100$.

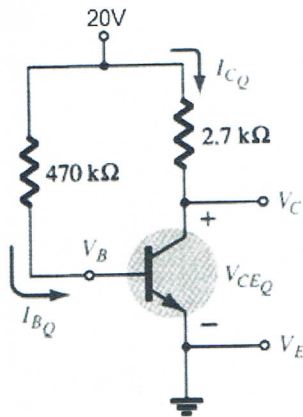


Figure 10.

4. Use this table below to determine the change in I_C from 100°C to 175°C for the transistor defined in this table for emitter-bias with $R_B = 240 \text{ k}\Omega$, $R_E = 2 \text{ k}\Omega$ and $\beta = 100$ due to the $S(V_{BE})$ stability factor.

$$S(V_{BE}) = \frac{-\beta}{R_B + (\beta + 1)R_E}$$

$T (^{\circ}\text{C})$	$I_{CO} \text{ (nA)}$	β	$V_{BE} \text{ (V)}$
-65	0.2×10^{-3}	20	0.85
25	0.1	50	0.65
100	20	80	0.48
175	3.3×10^3	120	0.3

5. Determine I_C , V_E and R_1 from Figure 11.

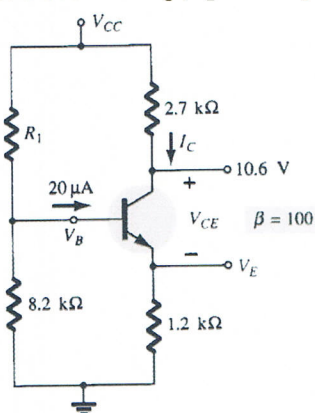
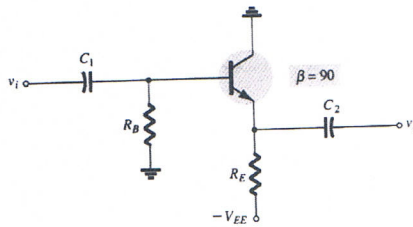


Figure 11.

Handout

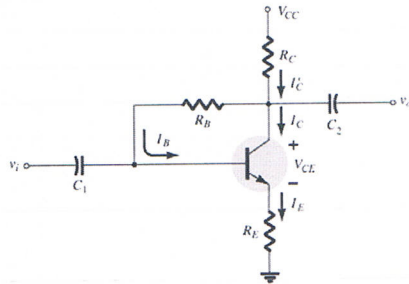
Emitter-follower



$$V_{CE} = V_{EE} - I_E R_E$$

$$I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E}$$

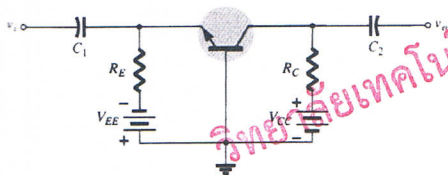
Collector-feedback



$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)}$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

Common-base



$$I_E = \frac{V_{EE} - V_{BE}}{R_E}$$

$$V_{CE} = V_{EE} + V_{CC} - I_E(R_C + R_E)$$

Class A amplifier

$$P_i(DC) = \frac{V_{CC}^2}{2R_C}$$

$$P_o(AC) = \frac{V_{CC}^2}{8R_C}$$

Transformer-couple class A amplifier

$$\%h = 50 \left(\frac{V_{CE(max)} - V_{CE(min)}}{V_{CE(max)} + V_{CE(min)}} \right)^2$$

$$\frac{R_L}{R_L} = \frac{R_1}{R_2} = \left(\frac{N_1}{N_2} \right)^2 = a^2$$

Class B amplifier

$$P_o(AC) = \frac{V_L^2(p-p)}{8R_L} = \frac{V_L^2(p)}{2R_L}$$

$$P_i(DC) = V_{CC} I_{dc} = V_{CC} \left(\frac{2}{\pi} I_L(p) \right)$$

Answer Sheet

Name: _____ ID: _____

Subject: 353152 ElectronicCircuits I

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